

$$\alpha - 1 = n \quad , \quad \beta - 1 = (n - k)$$

Stat 465 HW 2

2. $\frac{18}{21} \Rightarrow$ success, $w =$ probability of treatment being effective

Prior A: $w \sim \text{Beta}(1, 1)$

Prior B: $w \sim \text{Beta}(4, 0.9)$

(a) $w^{\alpha-1} (1-w)^{\beta-1}$

$$w^{18} (1-w)^3 w^{1-1} (1-w)^{1-1}$$

$$= w^{19-1} (1-w)^{4-1}$$

$$\boxed{w | X \sim \text{Beta}(19, 4)}$$

post = likelihood $\times$ prior

(b) $w^3 (1-w)^{-0.1}$

$$\pi_{13}(w|X) \propto w^{18} (1-w)^3 w^{4-1} (1-w)^{0.9-1}$$

$$= w^{21} (1-w)^{2.9}$$

$$= w^{22-1} (1-w)^{3.9-1}$$

$$\boxed{w | X \sim \text{Beta}(22, 3.9)}$$

(c)

$$w | X \sim \text{Beta}(\alpha + x, \beta + n - x)$$

$$\rightarrow \frac{\alpha}{(\alpha + \beta)}$$

$$E[w|X] = \frac{\alpha + x}{(\alpha + x) + (\beta + n - x)} = \frac{\alpha + x}{\alpha + \beta + n}$$

$$= \frac{\alpha}{\alpha + \beta + n} + \frac{x}{\alpha + \beta + n} = \left[\frac{\alpha + \beta}{\alpha + \beta + n} \cdot \frac{\alpha}{\alpha + \beta} + \frac{n}{\alpha + \beta + n} \cdot \frac{x}{n} \right]$$

$\frac{\alpha}{\alpha+\beta} \rightarrow$ prior mean, $\frac{x}{n}$ is the observed success proportion

$\hookrightarrow$ weights

$\hookrightarrow \frac{\alpha+\beta}{\alpha+\beta+1}$

$\hookrightarrow$ weights

$\hookrightarrow \frac{n}{\alpha+\beta+1}$

$$(d) E[W|x] = \frac{\alpha}{\alpha+\beta}$$

$$E[W] = \frac{19}{19+4} = \frac{19}{23} \approx \boxed{0.8261}$$

$\nearrow$
Beta(19, 4)

Beta(22, 3.9)

$$E[W] = \frac{22}{25.9} \approx \boxed{0.8494}$$

we see they differ since the prior
has a lower mean which means
prior A will pull the posterior mean
towards below the observed success
proportion

(2e) μ vs core

3.

$$w | x \sim \text{Beta}(22, 4)$$

(a) core in μ

$$(0.68781, 0.95462)$$

(b) given the observed data, and the assumed model and error, the posterior probability that w lies between 0.68781 and 0.95462 is 95%

$$(c) [0.70898, 0.96719]$$

(d) the shorter one is HPD with a smaller width

(e) $\text{Beta}(22, 4)$ is left skewed with a longer tail toward smaller values, equal tailed intervals meet = prob at both tails but HPD errorized highest-density values also HPD shifting toward larger values

4. $x = \text{group 1} = \frac{437}{980}$, $y = \text{group 2} = \frac{398}{820} = w_y$

$w_x \sim \text{Beta}(1,1)$, $w_y \sim \text{Beta}(1,1)$

(a) $L(w_x | x) = \binom{980}{437}$

$= w_x^{437} (1-w_x)^{543} \propto w_x^{437} (1-w_x)^{543}$

$\pi(w_x) \propto w_x^{1-1} (1-w_x)^{1-1} = 1$

$\pi(w_x | x) \propto w_x^{437} (1-w_x)^{543}$

$\alpha - 1 = 437$

$\beta - 1 = 543$

$\alpha = 438$

$\beta = 544$

$w_x | x \sim \text{Beta}(438, 544)$

(2) $\pi(w_y | y) \propto w_y^{398} (1-w_y)^{422} \propto w_y^{398} (1-w_y)^{422}$

$\alpha - 1 = 398$

$\beta - 1 = 422$

$\alpha = 399$

$\beta = 423$

$w_y | y \sim \text{Beta}(399, 423)$

(b)

$$E[L_x | X] = \frac{a}{a+b} = \frac{438}{438+544} \approx \boxed{0.4460}$$

$$E[L_y | Y] = \frac{399}{822} \approx \boxed{0.4854}$$

(c) $P(L_x < L_y | x, y)$

(d) If ^{population} the probability the female birth proportion for group 2 is greater than female birth proportion for group 2 given x, y

(e) the p-value is relative only gives the probability assuming the 2 population groups are equal. that observed difference at least as extreme as the one observed in the sample of group 1 having a lower proportion. but part (c) is the probability that group 1's female birth prop is lower than group 2's given observed data and prior

$$\theta = a + \sum_{i=1}^b x_i \quad / \quad b+1$$

5.

$$n=20 \quad / \quad \sum_{i=1}^{20} x_i = 54$$

$$\lambda \sim g_a(a=2, b=1)$$

$$(a) \quad L(\lambda | x) \propto \lambda^{\sum x_i} e^{-n\lambda}$$

$$L(\lambda | x) = \prod \frac{\lambda^{x_i} e^{-\lambda}}{x_i!} = \frac{\lambda^{\sum x_i} e^{-n\lambda}}{\prod x_i!}$$

$$L(\lambda | x) \propto \lambda^{54} e^{-20\lambda}$$

$$(b) \quad \pi(\lambda) \propto \lambda^{a-1} e^{-b\lambda} = \lambda^1 e^{-\lambda}$$

$$\pi(\lambda | x) \propto \lambda^{54+2-1} e^{-(20+1)\lambda}$$

$$\lambda | x \sim \text{gamma}(56, 21)$$

$$(c) \quad E[\lambda | x] = \frac{56}{21} \approx 2.667$$

$$[2.014371, 3.409050]$$

Q1) we know $x_i | \lambda \sim \text{Poisson}(\lambda)$
 $\hookrightarrow E[x_i | \lambda] = \lambda$

$$E[x_i | x] = E_{\lambda|x}[E[x_i | \lambda, x]] = E_{\lambda|x}[\lambda]$$

$$E[x_i | x] = \frac{56}{21} \approx 2.667$$

Given our λ , future counts will follow a Poisson distribution with a mean of λ and

Since we average the conditional Poisson mean, over the posterior distribution of λ .

$$Q2) \hat{P}(x_i \geq 4 | x) = 0.283$$

Given the observed data, there is approx 28.3% chance of receiving of at least 4 complaints on a future day.